

The Economy

Seminar 4: Social Interactions and Institutions

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Why models?

Social scientists are interested in **causal explanations** for social phenomena.

- A non-social example: lightning → thunder
- We seek the **causal mechanisms** behind empirical regularities
- Social examples abound:
 - economic downturn → incumbent loses election
 - democracies are less likely than autocracies to go to war with each other?

Underlying many causal arguments about social outcomes are **strategic interactions**.

- Goal-oriented people, rationally pursuing their own goals, engaging in conflict and cooperation with each other

What political questions can we ask with models?

International relations assumes states are unitary rational actors seeking survival. **Why does North Korea pursue nuclear weapons?**

American politics debates whether rational politicians are office motivated or policy motivated. **Why do politicians run for office? Why do voters vote even though one vote is not likely to change electoral outcomes?**

Comparative politics considers how a rational dictator trades off the incentives to extract rents from being leader and the risk of revolt. **Under what conditions can citizens solve their collective action problems and revolt against their leader?**

Political economics consider how a rational consumer chooses labor and leisure facing various tax regimes. **What is the optimal tax rate?**

Elements of game theory

- What agents play the game (**players**)?
- What agents can do (**strategies**)?
- What is the structure of the problem (**game form**)?
- What are agents' preferences over outcomes (**payoffs**)?
- What do the agents know (**information**)?

Bridging between preferences and game theory

We are ready to move from **individual preferences** to how people **act** on their preferences in strategic interactions with each other.

We start with **static games** of **complete information**:

- **Static games** take the following form:
 - ① the players simultaneously choose actions
 - ② they receive payoffs that depend on the combination of actions just chosen
- **Complete information** means everyone knows how every set of actions maps to outcomes

What we will learn

In this class, we will learn:

- ➊ How to translate a verbal statement of a strategic interaction to a formal game
 - **Normal-form representation** of a game
- ➋ What constitutes the “solution” to a game (**equilibrium**)
- ➌ How to solve for the equilibrium of a game
 - Iterated elimination of strictly dominated strategies
 - **Nash equilibrium**

Classic example: The Prisoners' Dilemma

By way of example, let's begin with the classic **Prisoners' Dilemma**, which has the following verbal statement:

- Two suspects are arrested and charged with a crime. The police lack sufficient evidence to convict them unless one of them confesses.
- The police take the two suspects to separate rooms and offer each of them the same proposition:
 - If nobody confesses, both will be convicted of a minor offense and sentenced to 1 month in jail
 - If one confesses, the one who confessed is released immediately and the other receives a 9-month sentence
 - If both confess, they each serve 6 months

Elements of a game

- A set of **players** $I = \{1, 2, \dots, n\}$, with an arbitrary player i
- A **strategy space** S_i which is the set of **actions** available to each player, where $s_i \in S_i$ is one action a player might take
- For each player, **preferences** over the set of outcomes captured by utility functions: $\{u_1, u_2, \dots, u_n\}$
 - where an “outcome” is fully specified by the strategies played by all players, i.e. $u_1(s_1, s_2, \dots, s_n)$
- In general, we also specify players’ **information sets**: for now, we assume “common knowledge”
- We also want to know the sequence of the game: for now, we have simultaneous play at one moment in time

The normal-form representation of a game

Definition (Normal-Form Representation)

The normal-form representation of an n -player game specifies:

- ① the players' strategy spaces $S_1, S_2, \dots, S_n$
- ② their payoff functions $u_1, u_2, \dots, u_n$

We denote this game by $G = S_1, S_2, \dots, S_n; u_1, u_2, \dots, u_n$.

Formalizing the Prisoners' Dilemma

Example (Prisoners' Dilemma)

		Clyde	
		Don't Confess	Confess
Bonnie	Don't Confess	-1,-1	-9,0
	Confess	0,-9	-6,-6

- Players: Bonnie (1, row player) and Clyde (2, column player)
- Strategies (Actions): $S_1 = \{DC, C\}$ and $S_2 = \{DC, C\}$
- Payoffs: numbers (utility representations) in the matrix above
- e.g. $u_1(s_1 = DC, s_2 = DC) = -1$

Strictly dominated strategies

At the very least, a rational player can rule out actions that don't yield the best outcomes **whatever the other players might do**.

Definition (Strictly Dominated Strategy)

In the normal-form game $G = S_1, \dots, S_n; u_1, \dots, u_n$, let s'_i and s''_i be feasible strategies, i.e. members of S_i .

Strategy s'_i is **strictly dominated** by strategy s''_i IF for each feasible combination of the other players' strategies, i 's payoff from playing s'_i is **strictly less** than i 's payoff from playing s''_i :

$$u_i(s_1, \dots, s_{i-1}, s'_i, s_{i+1}, \dots, s_n) < u_i(s_1, \dots, s_{i-1}, s''_i, s_{i+1}, \dots, s_n)$$

for each $(s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n)$ that can be constructed from the other players' strategy spaces $S_1, \dots, S_{i-1}, S_{i+1}, \dots, S_n$.

Does the Prisoners' Dilemma have strictly dominated strategies?

Iterated elimination of strictly dominated strategies

Does the Prisoners' Dilemma have strictly dominated strategies?

Example (Prisoners' Dilemma)

		Clyde	
		Don't Confess	Confess
Bonnie	Don't Confess	-1,-1	-9,0
	Confess	0,-9	-6,-6

- Let's look at Bonnie's decision; the game is symmetric, so the same goes for Clyde
- If Clyde Doesn't Confess: C (0) is strictly better than DC (-1)
- If Clyde Confesses: C (-6) is strictly better than DC (-9)
- So we have found a **strictly dominated strategy**: DC (or dominant strategy: C)

Some important notes about the Prisoners' Dilemma

Example (Prisoners' Dilemma)

		Clyde	
		Don't Confess	Confess
Bonnie	Don't Confess	-1,-1	-9,0
	Confess	0,-9	-6,-6

We have just solved for our first **equilibrium**. What did we learn?

- The equilibrium (**C**, **C**) is rational, but not socially optimal
- The solution would have been the same with any numbers that preserved the structure: Temptation > Reward > Punishment > Sucker ($0 > -1 > -6 > -9$), a reminder about ordinal utilities
- This structure is common to many social problems: price collusion, pandering, nuclear proliferation

What if there are no SDS?

Example

		P2		
		L	C	R
P1	T	0,4	4,0	5,3
	M	4,0	0,4	5,3
	B	3,5	3,5	6,6

We can quickly see that we cannot eliminate any strategies:

- Player 1: **T** best when **C**, **M** best when **L**, **B** best when **R**
- Player 2: **L** best when **T**, **C** best when **M**, **R** best when **B**

(If it's the best in some circumstance, it can't be strictly dominated.)

Nash equilibrium

We define this equilibrium concept by what it looks like when you're there (and return later to the matter of *how* we get there):

Definition (Nash Equilibrium)

In the normal-form game $G = S_1, \dots, S_n; u_1, \dots, u_n$, the strategies $(s_1^*, \dots, s_n^*)$ are **Nash equilibria** if, for each player i , s_i^* is (at least tied for) player i 's best response to the strategies specified for the $n-1$ other players, $(s_1^*, \dots, s_{i-1}^*, s_{i+1}^*, \dots, s_n^*)$:

$$u_i(s_1^*, \dots, s_{i-1}^*, s_i^*, s_{i+1}^*, \dots, s_n^*) \geq u_i(s_1^*, \dots, s_{i-1}^*, s_i, s_{i+1}^*, \dots, s_n^*)$$

for every feasible strategy $s_i \in S$. In other words, s_i^* solves:

$$\max_{s_i \in S} u_i(s_1^*, \dots, s_{i-1}^*, s_i, s_{i+1}^*, \dots, s_n^*)$$

Leaving aside the thorny issue of how the *players* got there, let's see how an analyst might get there (let's solve for a N.E.).

Strategy:

- ① We'll first need to identify Player 1's **best response** to every action Player 2 might take
- ② Then we identify Player 2's best response to every action Player 1 might take
- ③ Then we'll look for any cells where **both players** are simultaneously best-responding

Solving for NE in a Normal Form Game

Let's revisit the game form that we failed to solve using iterated elimination of strictly dominated strategies.

Example

		P2		
		L	C	R
P1	T	0,4	4,0	5,3
	M	4,0	0,4	5,3
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- 1 Start with Player 1

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② Move on to Player 2

Solving for NE in a Normal Form Game

Let's revisit the game form that we failed to solve using iterated elimination of strictly dominated strategies.

Example

		P2		
		L	C	R
P1	T	0,4	4,0	5,3
	M	4,0	0,4	5,3
	B	3,5	3,5	6,6

- The only cell where **both players** are simultaneously best-responding to one another is **(B,R)** and that is our only Nash equilibrium.

Multiple Nash equilibria

Let's look at another classic example below. Two hunters are out hunting for game. They can cooperate and hunt the stag or they can each catch a hare. Stag is more valuable but requires cooperation.

Example

		Hunter 2	
		Stag	Hare
Hunter 1	Stag	3,3	0,2
	Hare	2,0	1,1

- Find the Nash equilibrium / equilibria.

Multiple Nash equilibria

Example

		Hunter 2	
		Stag	Hare
Hunter 1	Stag	3,3	0,2
	Hare	2,0	1,1

- We have two Nash equilibria: **(S,S)** and **(H,H)**.
- While **(S,S)** is obvious, **(H,H)** is counter-intuitive.
- Nash equilibria are not always optimal but they are inherently stable.
No one regrets their decision.

Pareto efficiency

Definition (Pareto Efficiency)

A feasible combination of decisions for a collection of agents is said to be **Pareto efficient** if there does not exist another feasible combination of decisions under which each agent is at least as well off and some agent is strictly better off.

no other strategy profile s' such that $u_i(s') \geq u_i(s)$ for every player i and $u_j(s') > u_j(s)$ for some player j .

Does the worker-employer game below have a Pareto efficient strategy?

Example

		Employer	
		W	S
Worker	W	4,4	1,6
	S	6,1	2,2

PS4 Exercise 1. Advertising and the Prisoner's Dilemma (Part 1)

Consider two companies sharing the soap market. They both have the choice of whether or not to launch an advertising campaign. The size of the soap market is not affected by any advertising campaign, but running an advertising campaign allows to gain market share.

The following table shows the profits of the two companies in millions of euros. The first number in brackets corresponds to the profit of firm A and the second to that of firm B.

A/B	No Advertising	Advertising
No Advertising	10,5	7,6
Advertising	13,-2	8,1

- ❶ Is there a dominant strategy for firm A? For firm B?

PS4 Exercise 1. Advertising and the Prisoner's Dilemma (Part 2)

A/B	No Advertising	Advertising
No Advertising	10,5	7,6
Advertising	13,-2	8,1

- 2 Is there a Nash equilibrium in this example? Is it Pareto optimal?
- 3 To carry out this project, firm A proposes to buy firm B and is willing to pay a price higher than firm B's current price value. How might this purchase be a good investment?

Finitely repeated games: Prisoners' Dilemma

- One of the most studied games is the repeated prisoner's dilemma.
- Consider an application focused on trade policy.
- Suppose the world economy performs better when all nations agree to free trade, but that individual countries prefer to protect their domestic economy.
- Given this tension, how are free trade regimes sustained?
- One answer is that free trade can be supported as an equilibrium in a repeated game where a trade war begins whenever a major country defects from the trade agreement.

Finitely repeated games: Prisoners' Dilemma

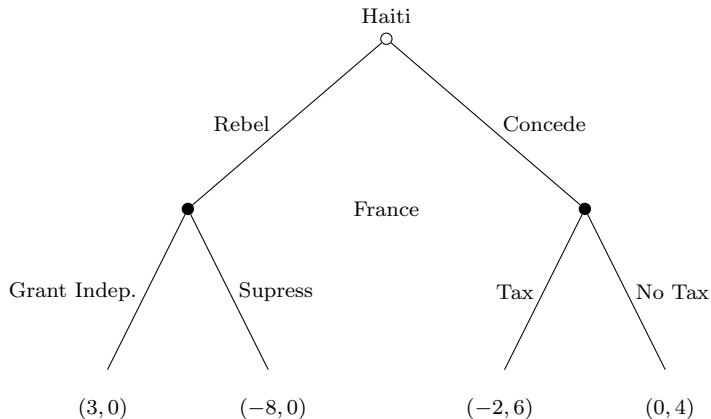
		EU	
		Free Trade	Protect
US	Free Trade	10,10	1,12*
	Protect	12*,1	4*,4*

- When the game is played once, the unique Nash equilibrium is the strategy profile (**Protect**, **Protect**).
- If the EU chooses to always **Free Trade**, then the best option of the US is to **Protect**.

Because the Prisoner's Dilemma has only one Nash equilibrium, it is impossible to encourage cooperation with the promise of coordinating on a good equilibrium or the threat of coordinating on a bad equilibrium.

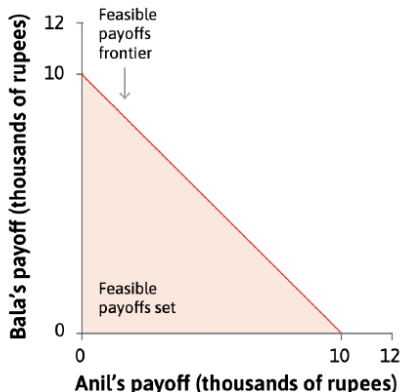
If the stage game G has a unique Nash equilibrium then, for any finite T , the repeated game $G(T)$ has a unique subgame-perfect outcome: the Nash equilibrium of G is played in every stage.

Extensive-Form Games



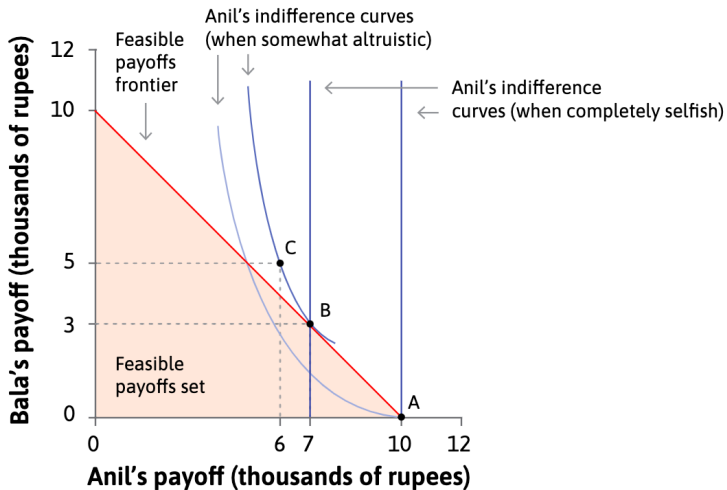
- Dynamic games where the decisions are taken with order (unlike simultaneously as before), can be represented in an extensive-form.
- Sometimes we are interested in dynamics: how incentives change as players learn where they are in a game.

PS4 Exercise 2. Altruism (Part 1)

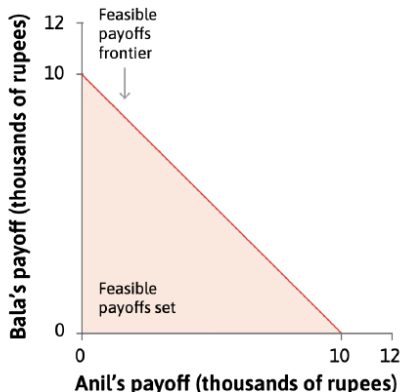


- 1 What would Anil's indifference curves look like if Anil's utility only increases if his consumption increases proportionally with Bala?
- 2 What would they look like if he derived utility only from the total sum of his consumption and the one of Bala?

PS4 Exercise 2. Altruism (solutions)



PS4 Exercise 2. Altruism (Part 2)



- What would it look like if he got utility only from his own consumption / if he got utility only from Bala's consumption?
- In each of these cases, propose a real-world situation in which Anil might have these preferences, noting how Anil and Bala get their gains.

- **PS5 Exercise 2** due in class.
- **Quiz 3** for next class!
- The midterm will cover Lectures 1-5.
- A mock midterm will be posted on Moodle this week.